# PAG7936LT: 1 Mega Pixel Global Shutter Color Image Sensor

## General Description

The PAG7936LT is an ultra-low-power global shutter color image sensor. It supports 1280 x 800 resolution and up to 120 frames per second (fps) image capture capability to output clear, low-noise images in low-light and bright scenes.

PAG7936LT's low-power architecture enables the always-on feature in the customer's applications, which is especially beneficial for the battery-powered system. The sensor is suitable for consumer products, machine vision, or industrial applications such as augmented reality (AR), virtual reality (VR), AI classification & recognition, collision avoidance, barcode applications, factory automation, etc.

PAG7936LT registers can be accessed through the standard $I^2C$ interface, while the image output is through MIPI or parallel interface.

## Key Features

- Ultra-low-power consumption
- LED Strobe
- RGB Lens shading compensation
- Programmable control for
  - Frame rate
  - Mirror & Flip
  - WOI
- Supports image resolution: 1280 x 800, 640 x400, 320 x 200
- Data format: 8-bit/10-bit RAW
- Supports $I^2C$ multi-slave ID by I/O strapping
- Auto Exposure and Gain Control
- De-noise
- Context Switch
- Fast AE

## Applications

- VR/AR
- Face Recognition
- Factory Automation
- Drone/ Barcode Reader

## Key Parameters

| Parameter | Value |
|---|---|
| Optical Format | 1/4” |
| Array Size | 1288 x 808 pixel |
| Pixel Size | 3.0 x 3.0 $\mu m^2$ |
| Max. CRA | 14.45 deg |
| Shutter Type | Global shutter |
| Max. Frame Rate | 120 fps |
| Signal to Noise Ratio, SNR | 36.9 dB |
| Dynamic Range | 66 dB |
| Sensitivity | 2751 mV/Lux-Sec at 530nm |
| Supply Voltage | VDDMA: 2.8V<br>VDD12: 1.2V<br>I/O: 1.8 to 2.8V |
| Power Consumption | 82.3mW (1280 x 800,120fps) |
| Operating Temperature, Tj | -40 to +85 °C |
| Stable image | 0 to +65 °C |
| Package Dimension | 5.050 x 5.470 x 0.768 $mm^3$ |

## Ordering Information

| Part Number | Description | Package Type | Packing Type |
|---|---|---|---|
| PAG7936LT | Global Shutter Image Sensor | Chip Scale Package | Chip Tray |

RoHS compliant

For any additional inquiries, please contact us at http://www.pixart.com

Preliminary: This product is still under product development where the product specifications are subjected to change upon product release.

# Table of Contents

## List of Figures

## List of Tables

# 1.0 Introduction

## 1.1 Overview

The architecture of PAG7936LT is based on CMOS image sensor technology. It is a global shutter image sensor with 1280 x 800 resolution and is designed for computer vision, or fast-moving object image applications to meet the growing market demand. The sensor can be controlled through the $I^2C$ interface and outputs the image data via the Parallel or MIPI interface.

The sensor supports partial array sensing such as flexible WOI region configuration, pixel average, or pixel skipping functions to reduce data transfer or power saving based on application needs.

**Note:** Throughout this document, the PAG7936LT is referred to as the "sensor".

## 1.2 Block Diagram

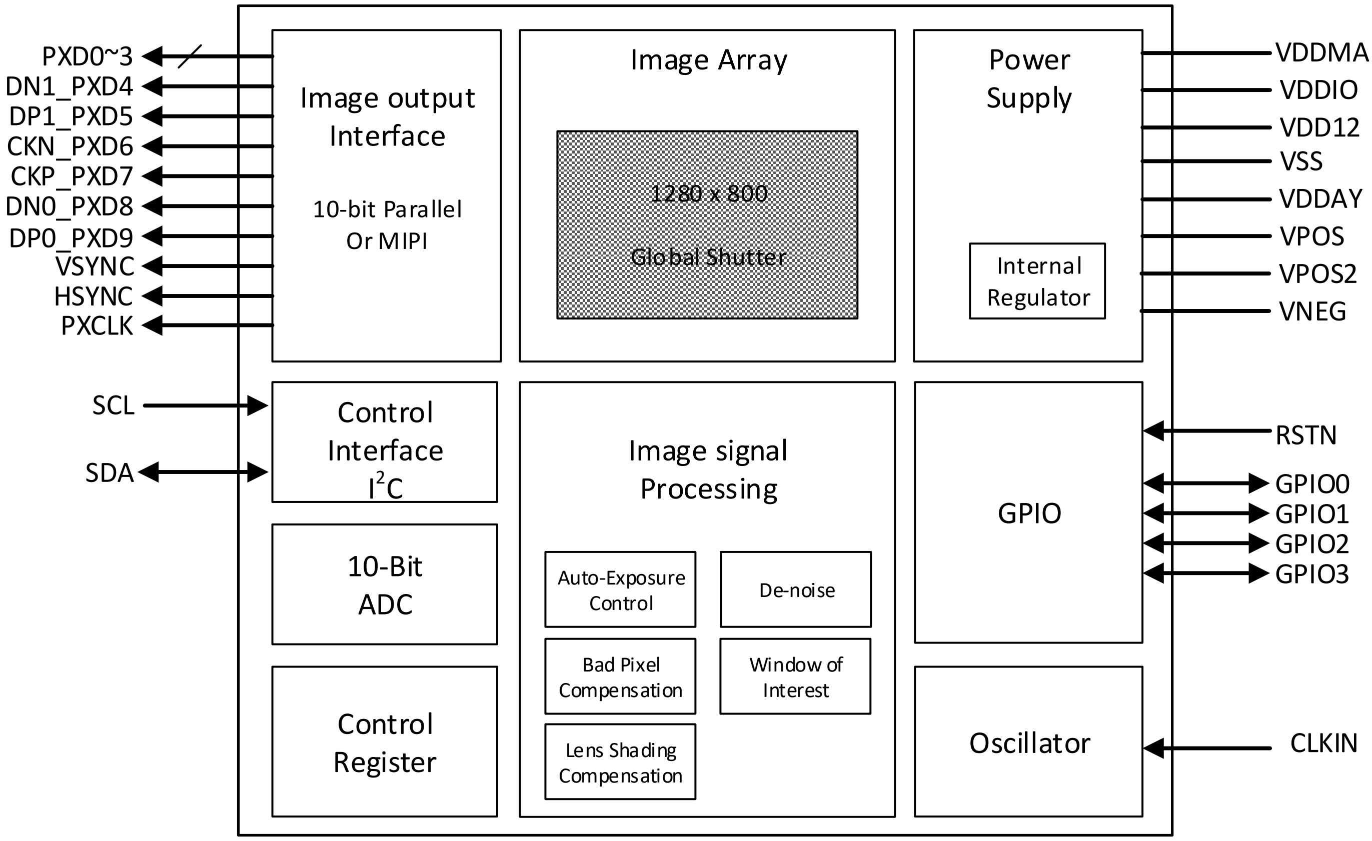


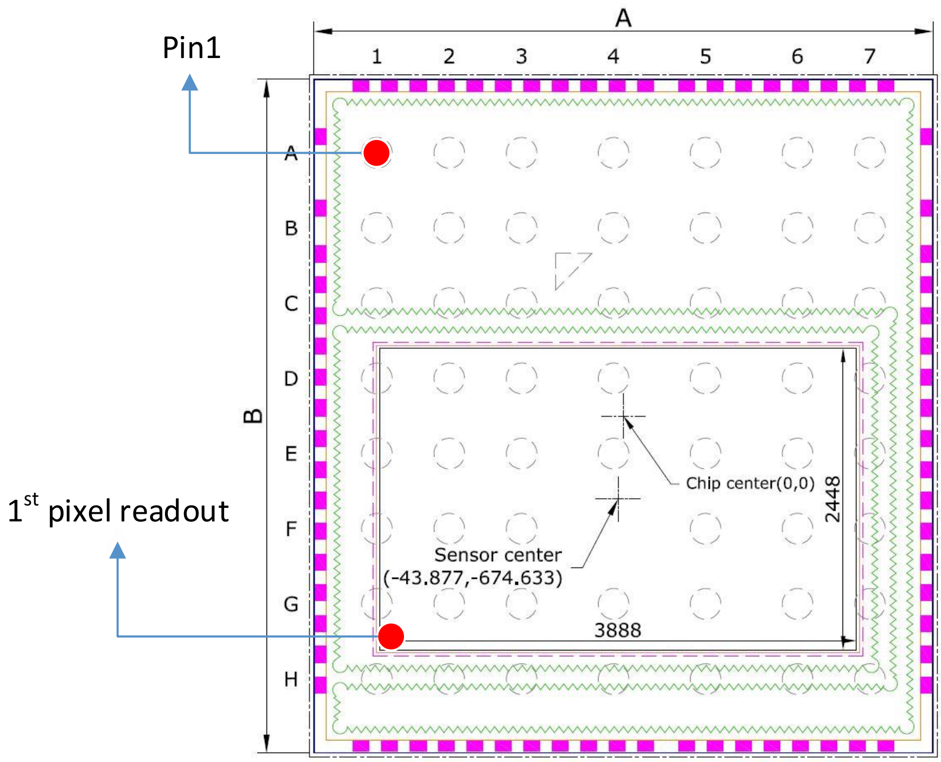

Figure 1. Block Diagram

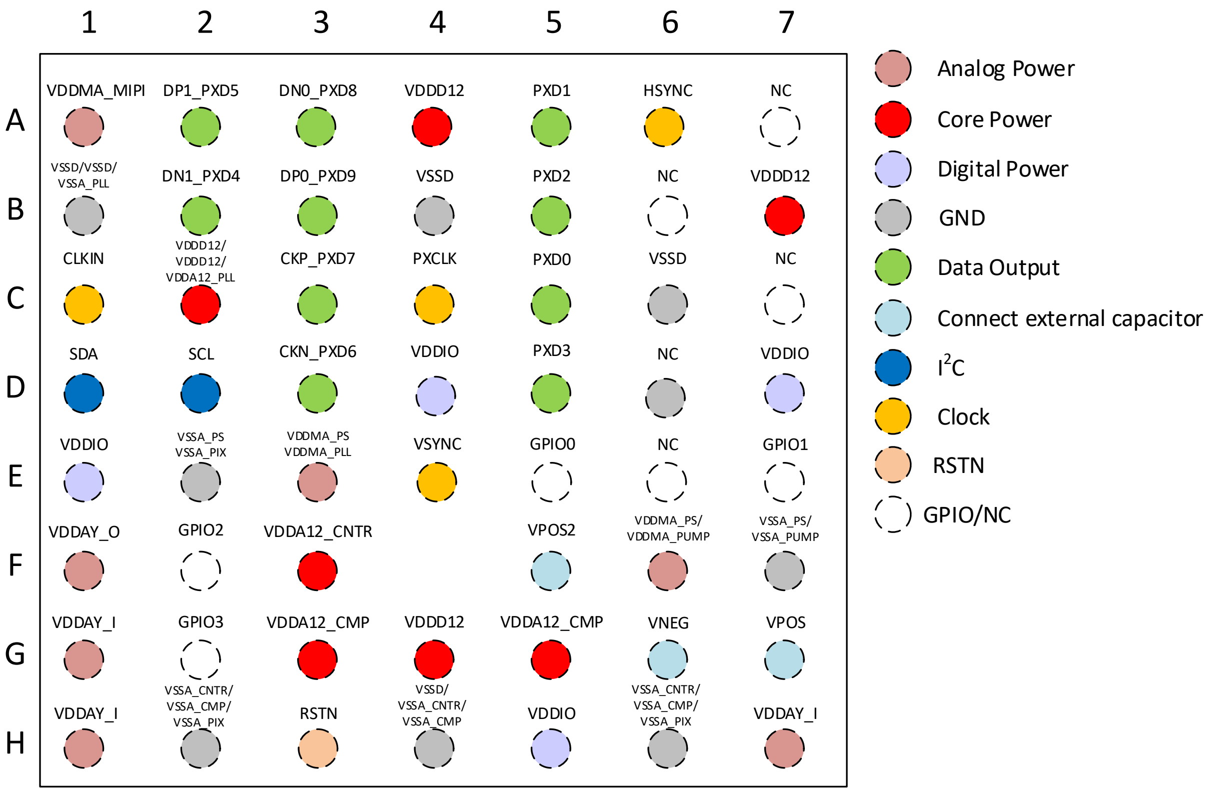
## 1.3 Terminology

Table 1. Terminology

| Term | Description |
|---|---|
| ISP | Image Signal Processing |
| $I^2C$ | Inter-Integrated Circuit |
| MIPI | Mobile Industry Processor Interface |
| Min. | Minimum |
| Typ. | Typical |
| Max. | Maximum |

## 1.4 Pin Assignment and Signal Description






| | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| **A** | VDDMA_MIPI | DP1_PXD5 | DN0_PXD8 | VDDD12 | PXD1 | HSYNC | NC |
| **B** | VSSD/VSSD/ VSSA_PLL | DN1_PXD4 | DP0_PXD9 | VSSD | PXD2 | NC | VDDD12 |
| **C** | CLKIN | VDDD12/ VDDD12/ VDDA12_PLL | CKP_PXD7 | PXCLK | PXD0 | VSSD | NC |
| **D** | SDA | SCL | CKN_PXD6 | VDDIO | PXD3 | NC | VDDIO |
| **E** | VDDIO | VSSA_PS/ VSSA_PIX | VDDMA_PS/ VDDMA_PLL | VSYNC | GPIO0 | NC | GPIO1 |
| **F** | VDDAY_O | GPIO2 | VDDA12_CNTR/ VDDA12_CNTR | | VPOS2 | VDDMA_PS/ VDDMA_PUMP | VSSA_PS/ VSSA_PUMP |
| **G** | VDDAY_I | GPIO3 | VDDA12_CMP/ VDDA12_CMP | VDDD12 | VDDA12_CMP | VNEG | VPOS |
| **H** | VDDAY_I | VSSA_CNTR/ VSSA_CMP/ VSSA_PIX | RSTN | VSSD/ VSSA_CNTR/ VSSA_CMP | VDDIO | VSSA_CNTR/ VSSA_CMP/ VSSA_PIX | VDDAY_I |

Figure 2. Pin Assignment (Top View)

Table 2. Signal Description

| Function | Pin No. | Signal Name | Type | Description |
|---|---|---|---|---|
| Power Supply | A4 | VDDD12 | Power | Core power supply |
| | B7 | VDDD12 | Power | |
| | C2 | VDDD12/ VDDD12 / VDDA12_PLL | Power | |
| | F3 | VDDA12_CNT/ VDDA12_CNTR | Power | |
| | G3 | VDDA12_CMP/ VDDA12_CMP | Power | |
| | G4 | VDDD12 | Power | |
| | G5 | VDDA12_CMP | Power | |
| | D7 | VDDIO | Power | I/O power supply |
| | E1 | VDDIO | Power | |
| | H5 | VDDIO | Power | |
| | A1 | VDDMA_MIPI | Power | Main power supply |
| | E3 | VDDMA_PLL / VDDMA_PS | Power | |
| | F6 | VDDMA_PS / VDDMA_PUMP | Power | |
| | F1 | VDDAY_O | Power | Internal regulator |
| | G1 | VDDAY_I | Power | connect to VDDAY_O |
| | H1 | VDDAY_I | Power | |
| | H7 | VDDAY_I | Power | |
| | G7 | VPOS | Power | Internal regulator |
| | F5 | VPOS2 | Power | Internal regulator |
| | G6 | VNEG | Power | Internal regulator |
| | B1 | VSSD / VSSD / VSSA_PLL | Ground | Reference Ground |
| | B4 | VSSD | Ground | |
| | C6 | VSSD / VSSD | Ground | |
| | E2 | VSSA_PS / VSSA_PIX | Ground | |
| | F7 | VSSA_PS / VSSA_PUMP | Ground | |
| | H2 | VSSA_PIX / VSSA_CMP / VSSA_CNTR | Ground | |
| $I^2C$ Interface | D2 | SCL | Input | $I^2C$ communications clock |
| | D1 | SDA | I/O | $I^2C$ communications data |
| MIPI or Parallel Data Interface | B3 | DP0_PXD9 | Output | MIPI Data Lane 0 or Pixel Data Output bit[9] of Parallel interface |
| | A3 | DN0_PXD8 | Output | MIPI Data Lane 0 or Pixel Data Output bit[8] of Parallel interface |

| Function | Pin No. | Signal Name | Type | Description |
|---|---|---|---|---|
| | C3 | CKP_PXD7 | Output | MIPI Clock Lane or Pixel Data Output bit[7] of Parallel interface |
| | D3 | CKN_PXD6 | Output | MIPI Clock Lane or Pixel Data Output bit[6] of Parallel interface |
| | A2 | DP1_PXD5 | Output | MIPI Data Lane 1 or Pixel Data Output bit[5] of Parallel interface |
| | B2 | DN1_PXD4 | Output | MIPI Data Lane 1 or Pixel Data Output bit[4] of Parallel interface |
| | A6 | HSYNC | Output | Line signal of parallel interface |
| | E4 | VSYNC | Output | Frame signal of parallel interface |
| | C4 | PXCLK | Output | Pixel clock of parallel interface |
| | C5 | PXD0 | Output | Pixel Data Output bit[0] of Parallel interface |
| | A5 | PXD1 | Output | Pixel Data Output bit[1] of Parallel interface |
| | B5 | PXD2 | Output | Pixel Data Output bit[2] of Parallel interface |
| | D5 | PXD3 | Output | Pixel Data Output bit[3] of Parallel interface |
| Functional I/O | H3 | RSTN | Input | Reset pin (Active low) |
| | C1 | CLKIN | Input | System clock input for oscillator |
| | E5 | GPIO0 | I/O | General-purpose I/O Pin |
| | E7 | GPIO1 | I/O | General-purpose I/O Pin |
| | F2 | GPIO2 | I/O | General-purpose I/O Pin |
| | G2 | GPIO3 | I/O | General-purpose I/O Pin |
| No connect | E6 | NC | N/A | No Connection, left floating |
| | D6 | NC | N/A | No Connection, left floating |
| | C7 | NC | N/A | No Connection, left floating |
| | A7 | NC | N/A | No Connection, left floating |
| | B6 | NC | N/A | No Connection, left floating |

## 1.5 Pixel Map

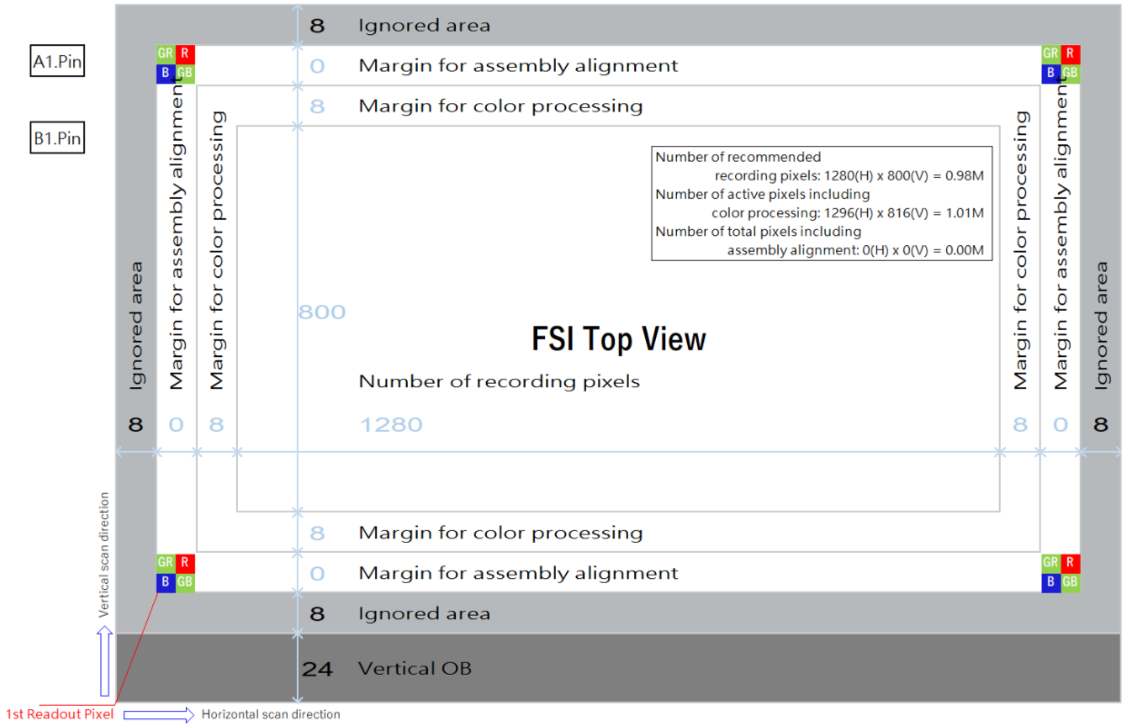



Figure 3. Pixel Array

# 2.0 Operating Specifications

## 2.1 Absolute Maximum Ratings

Table 3. Absolute Maximum Ratings

| Parameter | Symbol | Min. | Max. | Unit | Note |
|---|---|---|---|---|---|
| Storage Temperature | $T_S$ | | 125 | °C | |
| Main Voltage Supply 1 (VDDMA) | VDDMA_MIPI | -0.3 | 4.5 | V | |
| | VDDMA_PLL/ VDDMA_PS | -0.3 | 4.5 | V | |
| | VDDMA_PS/ VDDMA_PUMP | -0.3 | 4.5 | V | |
| Main Voltage Supply 2 (VDD12) | VDDD12 | -0.3 | 3 | V | |
| | VDDD12_CNTR | -0.3 | 3 | V | |
| I/O Voltage Supply | VDDIO | -0.3 | 4.5 | V | |
| I/O pin voltage | - | -0.3 | VDDIO+0.3 | V | |
| Relative Humidity | RH | 0 | 85 | % | Non-condensing, Non-biased |
| ESD | $ESD_{HBM}$ | - | 2 | kV | Class 2 on all pins, as per human body model. JESD22-A114E with 15 sec zap intervals. |
| Lead-free Solder Temperature | $T_P$ | - | 260 | °C | Refer to Assembly Guide |

Notes:

1. At ambient temperature = 25°C.
2. Maximum Ratings are the maximum parameter values that can damage the device when exceed this limit.
3. Exposure to these conditions or conditions beyond those indicated may adversely affect device reliability. Functional operation under absolute maximum-rated conditions is not implied.

## 2.2 Recommended Operating Conditions

Table 4. Recommended Operating Conditions

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Junction Temperature | Tj | -40 | 25 | 85 | °C | |
| Stable image | - | 0 | - | 50 | °C | |
| Main Voltage Supply 1 (VDDMA) | VDDMA_MIPI | 2.7 | 2.8 | 2.94 | V | Include ripples |
| | VDDMA_PLL/ VDDMA_PS | 2.7 | 2.8 | 2.94 | V | Include ripples |
| | VDDMA_PS/ VDDMA_PUMP | 2.7 | 2.8 | 2.94 | V | Include ripples |
| Main Voltage Supply 2 (VDD12) | VDDD12 | 1.14 | 1.2 | 1.26 | V | Include ripples |
| | VDDD12/ VDDD12/ VDDA12_PLL | 1.14 | 1.2 | 1.26 | V | Include ripples |
| I/O Voltage Supply | VDDIO | 1.62 | 2.8 | 2.94 | V | Include ripples |
| Digital I/O Driving Ability | $I_{DDIODR}$ | - | - | 16 | mA | At VDDIO= 2.8V |
| | $I_{DDIODR}$ | - | - | 12 | mA | At VDDIO= 1.8V |
| VDDMA Rise Time | $t_{R\_MA}$ | 0.1 | - | 2 | ms | 0 to minimum VDDMA |
| VDDD12 Rise Time | $t_{R\_12}$ | 0.1 | - | 2 | ms | 0 to minimum VDDD12 |
| VDDIO Rise Time | $t_{R\_IO}$ | 0.1 | - | 2 | ms | 0 to minimum VDDIO |
| EXT input clock frequency | $f_{extclk}$ | 6 | 24 | 27 | MHz | Square wave<br>Duty cycle= 45% to 55% |

**Note:** At ambient temperature = 25°C.

## 2.3 DC Electrical Characteristics

Table 5. DC Electrical Specifications

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Continuous mode Current at MIPI interface | $I_{DDMA}$ | - | 16 | - | mA | At 1280 x 800 image size, 120fps |
| | $I_{DDD12}$ | - | 31 | - | | At 1280 x 800 image size, 120fps (Bad Pixel Correction off) |
| | $I_{IO}$ | 100 | - | - | µA | In MIPI interface |
| Suspend State Current | $I_{DDMA}$ | - | 3 | - | µA | |
| | $I_{DDD12}$ | - | 30 | - | | |
| I/O Input High Voltage | $V_{IH}$ | 0.8x VDDIO | - | VDDIO +0.3 | V | |
| I/O Input Low Voltage | $V_{IL}$ | -0.3 | - | 0.2x VDDIO | V | |
| I/O Output High Voltage | $V_{OH}$ | 0.8x VDDIO | - | - | V | |
| I/O Output Low Voltage | $V_{OL}$ | - | - | 0.2x VDDIO | V | |

Notes:

1. At ambient temperature= 25°C.
2. All parameters are tested under operating conditions: VDDMA= 2.8V, VDDD12= 1.2V, VDDIO= 1.8V, $T_A$= 25°C.

## 2.4 AC Electrical Characteristics

Table 6. AC Electrical Specifications

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Switch to suspend | $t_{Stm}$ | - | - | 1 | fps | switch to suspend state needs to delay 1 frame period. |
| MIPI CSI-2 Data Speed | $f_{MIPI}$ | - | 800 | - | Mbit/s | |
| LPO Clock | $f_{LPO}$ | - | 1000 | - | kHz | |
| PXCLK and PXD rise time | $t_R$ | | 1.6 | 2.1 | ns | At $C_{load}$= 16pF and Slew rate= code 0 |
| PXCLK and PXD fall time | $t_F$ | | 1.6 | 2.1 | ns | At $C_{load}$= 16pF and Slew rate= code 0 |

**Note:** At ambient temperature = 25°C.

## 2.5 Chip Array Specifications

Table 7. Array Specifications

| Parameter | Symbol | Value | Unit | Note |
|---|---|---|---|---|
| Horizontal Image Array Count | - | 1280 | pixel | |
| Vertical Image Array Count | - | 800 | pixel | |
| Pixel Size | - | 3 x 3 | $\mu m^2$ | |
| CRA | - | < 14.45 | deg | |
| Signal to Noise Ratio | - | 36.9 | dB | |
| Dynamic Range | - | 66 | dB | |
| Sensitivity | - | 2751 | mV/Lux-Se | @530nm |

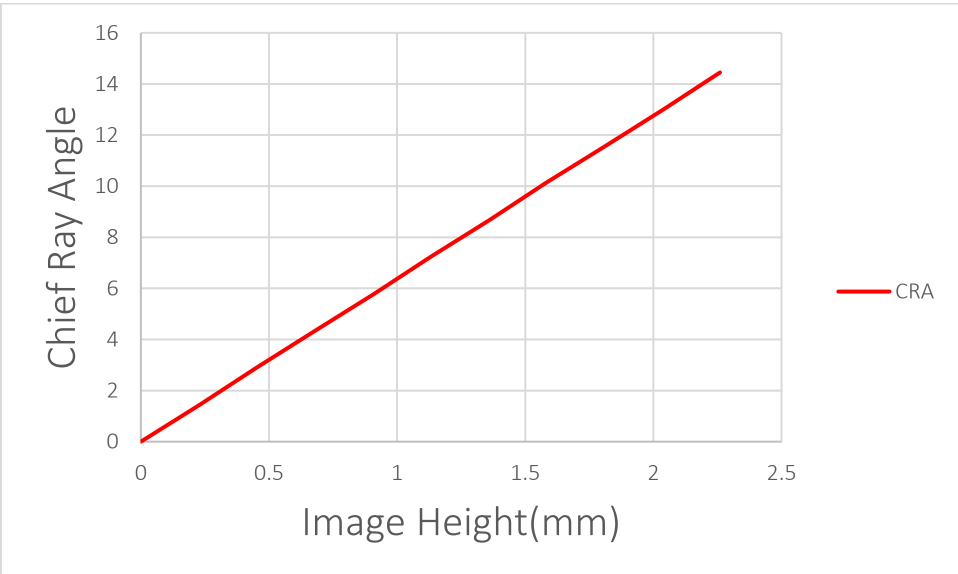



Figure 4. CRA Curve

Table 8. Chief Ray Angle

| Field | Image Height | CRA(Deg) |
|---|---|---|
| 0 | 0 | 0 |
| 0.1 | 0.23 | 1.45 |
| 0.2 | 0.45 | 2.89 |
| 0.3 | 0.68 | 4.34 |
| 0.4 | 0.91 | 5.78 |
| 0.5 | 1.13 | 7.23 |
| 0.6 | 1.36 | 8.67 |
| 0.7 | 1.58 | 10.12 |
| 0.8 | 1.81 | 11.56 |
| 0.9 | 2.04 | 13.01 |
| 1 | 2.26 | 14.45 |

## 2.6 Spectrum Response

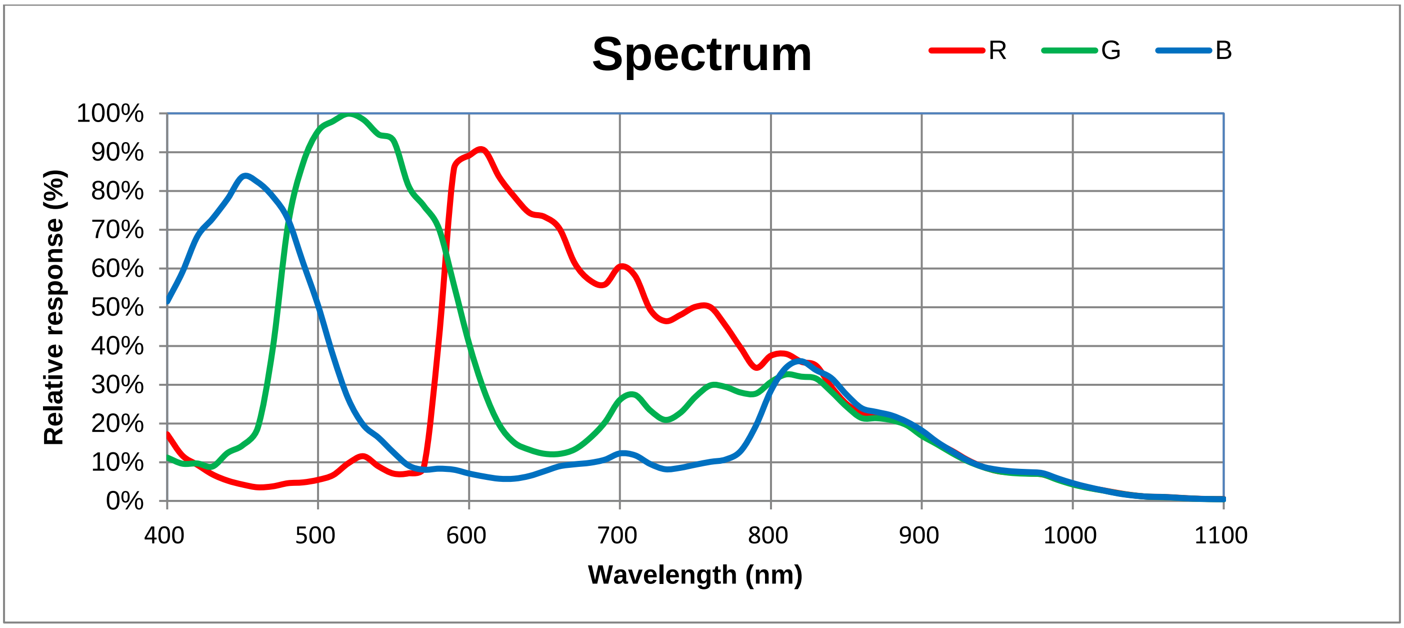



Figure 5. Spectrum Response Curve

# 3.0 Mechanical Specifications

## 3.1 Package Mechanical Dimension

| | Symbol | Nominal | Min. | Max. |
|---|---|---|---|---|
| | | | µm | |
| Package Body Dimension X | **A** | 5050 | 5025 | 5075 |
| Package Body Dimension Y | **B** | 5470 | 5445 | 5495 |
| Package Height | **C** | 768 | 708 | 828 |
| Ball Height | **C1** | 130 | 100 | 160 |
| Package Body Thickness | **C2** | 638 | 593 | 683 |
| Thickness of Glass surface to wafer | **C3** | 442 | 422 | 462 |
| Ball Diameter | **D** | 250 | 220 | 280 |
| Total Pin Count | **N** | 55 | | |
| Pin Count X axis | **N1** | 7 | | |
| Pin Count Y axis | **N2** | 8 | | |
| Pins Pitch X axis | **J1** | 590 | | |
| | **J3** | 750 | | |
| Pins Pitch Y axis | **J2** | 610 | | |
| Edge to Pin Center Distance along X | **S1** | 515 | 485 | 545 |
| Edge to Pin Center Distance along Y | **S2** | 600 | 570 | 630 |

＊1. The thickness of Bond-1 glass is 400um (Green glassx1pcs).
＊2. The material of solder printing is SAC305.
＊3. The Si thickness is 150um.

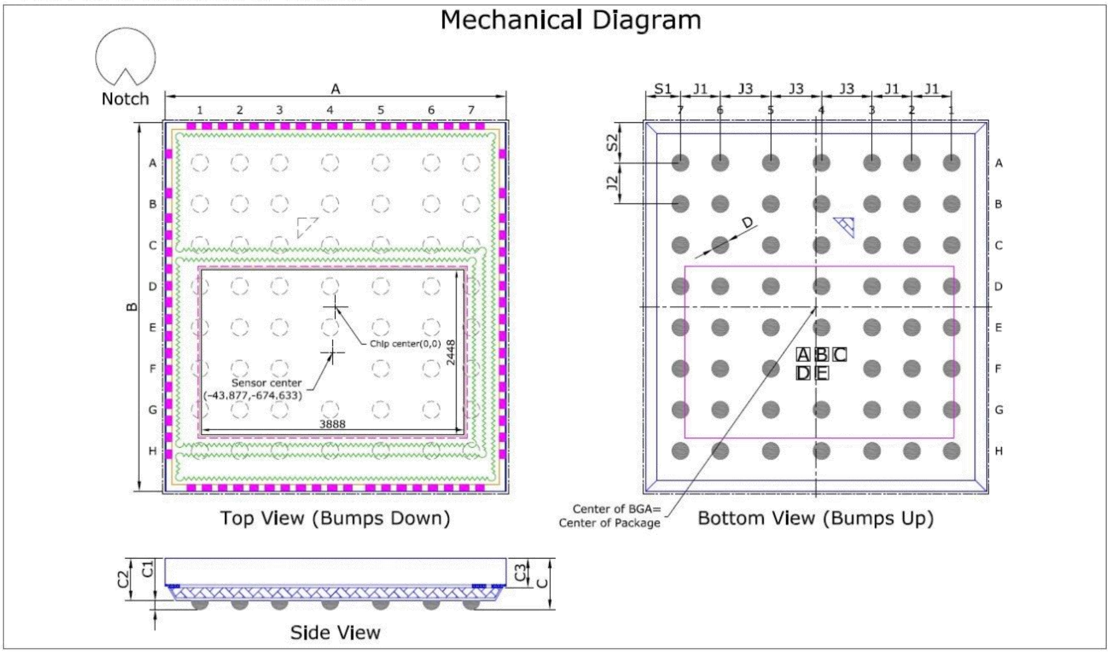



Figure 6. Package Outline Dimension

## 3.2 Package Marking Identification

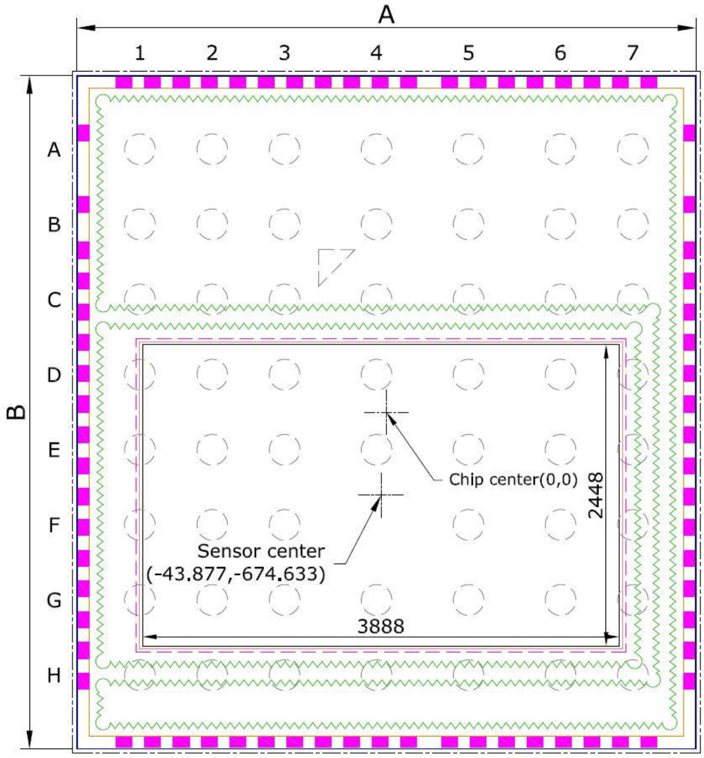
Refer to Figure 6 view for the code marking location on the device package.

Table 9. Code Identification

| Marking | Description |
|---|---|
| ABCD | AB= Date code |
| E | E= Wafer ID |

## 3.3 Image Array Dimension

Chip center and sensor center position refer to Figure 7.




Top View (Bumps Down)

Figure 7. 1st Pixel Location of Image Array

## 3.4 Packing Information

### 3.4.1 Carrier Drawing

Table 10. Packing Information

| Parameter | Quantity | Unit | Note |
|---|---|---|---|
| Sensors per tray | 42 | piece | |
| Trays per stack | 10 | tray | Add a top cover on top of each stack. Refer to Figure 8 |
| Stacks per packing bag | 2 | stack | Figure 9 |
| Packing bags per packing box | 5 | bag | Figure 10 |
| Max. sensors per packing box | 4200 | piece | |

Figure 8. Stack 10 +1 Chip Tray


Figure 9. Packing Bag

Figure 10. Packing Box

### 3.4.2 Unit Placement Orientation

IC CSP pin 1 (A1 pin) orientation is toward the chamfer of chip tray.

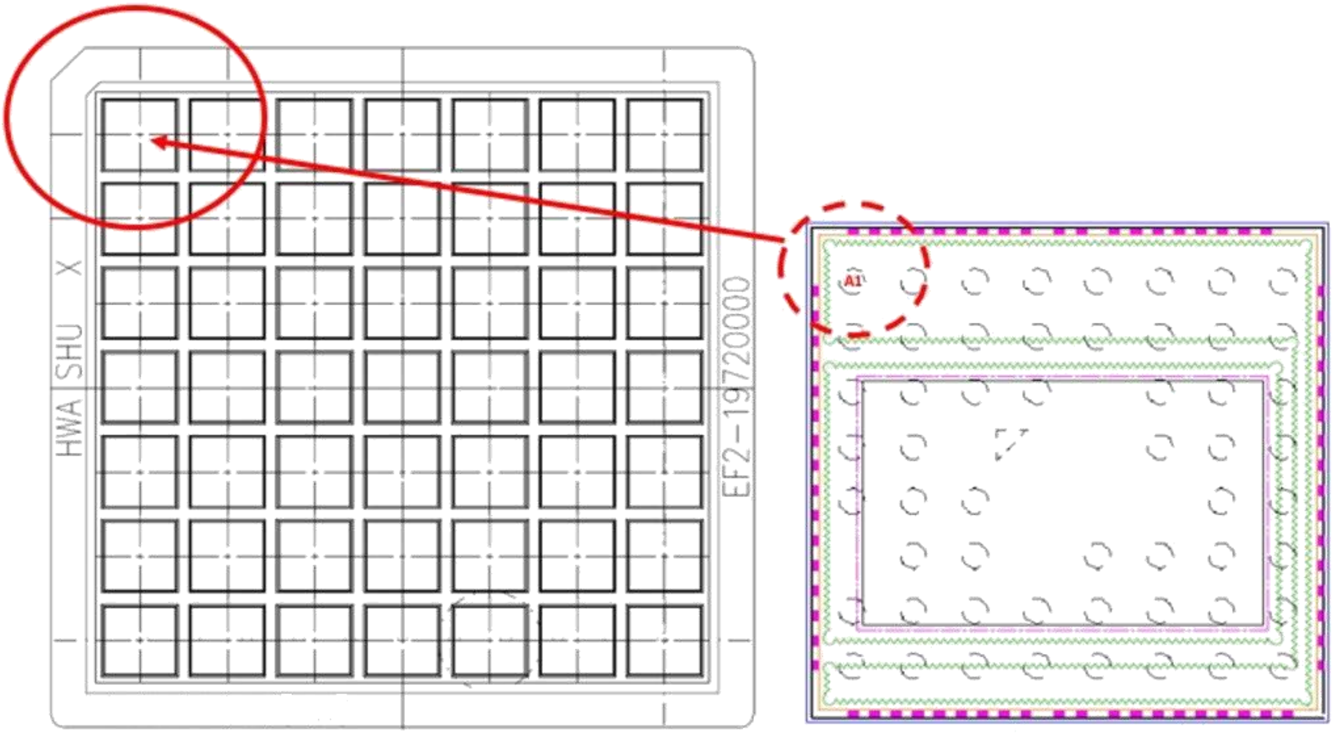
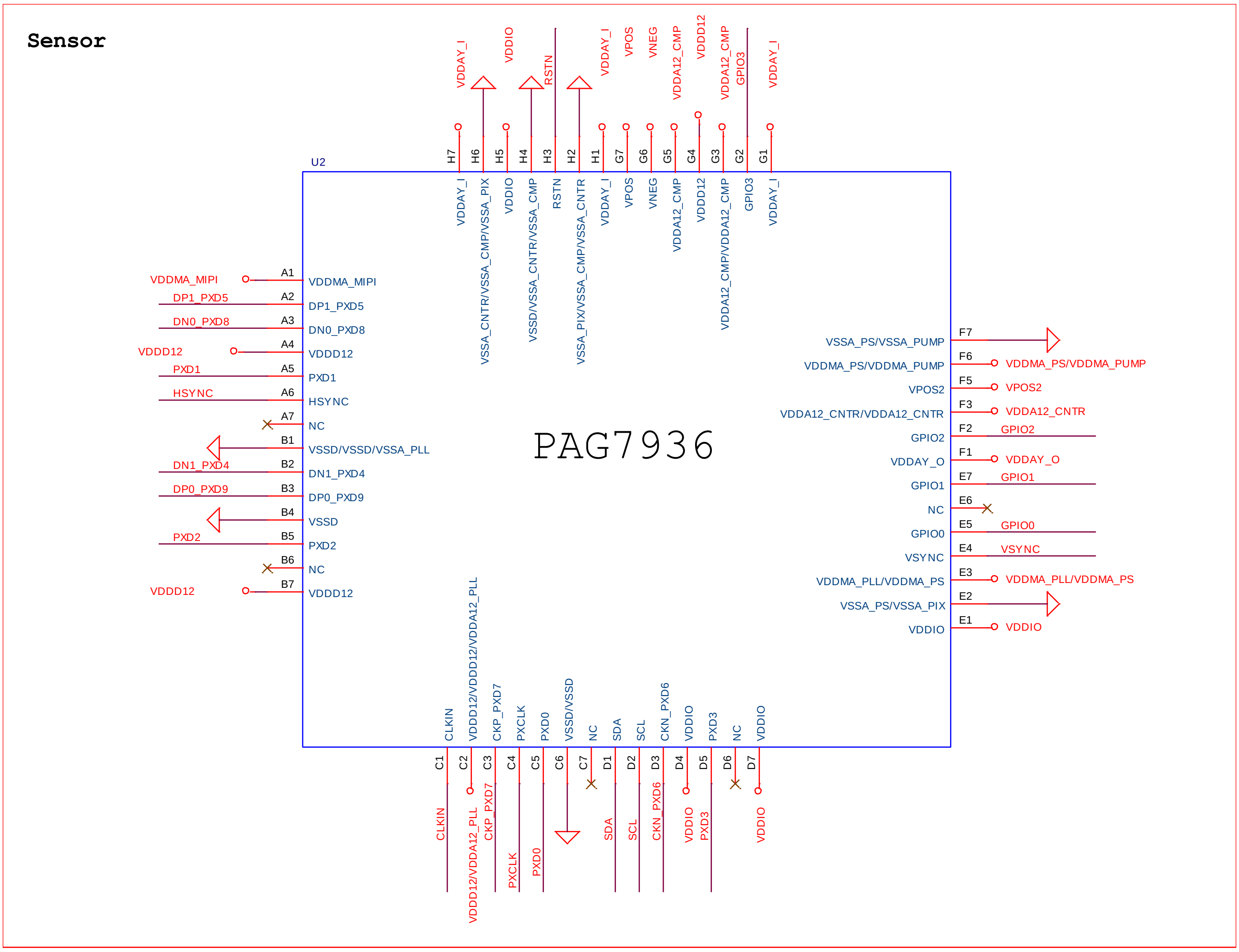
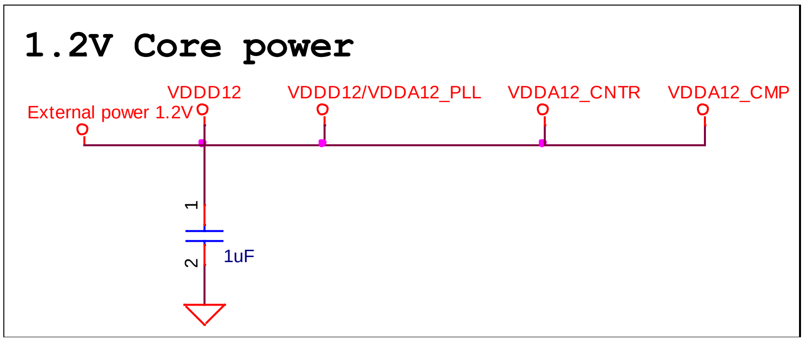


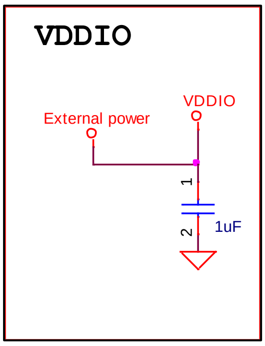

Figure 11. IC Orientation

# 4.0 Design Reference

## 4.1 General Reference Schematic



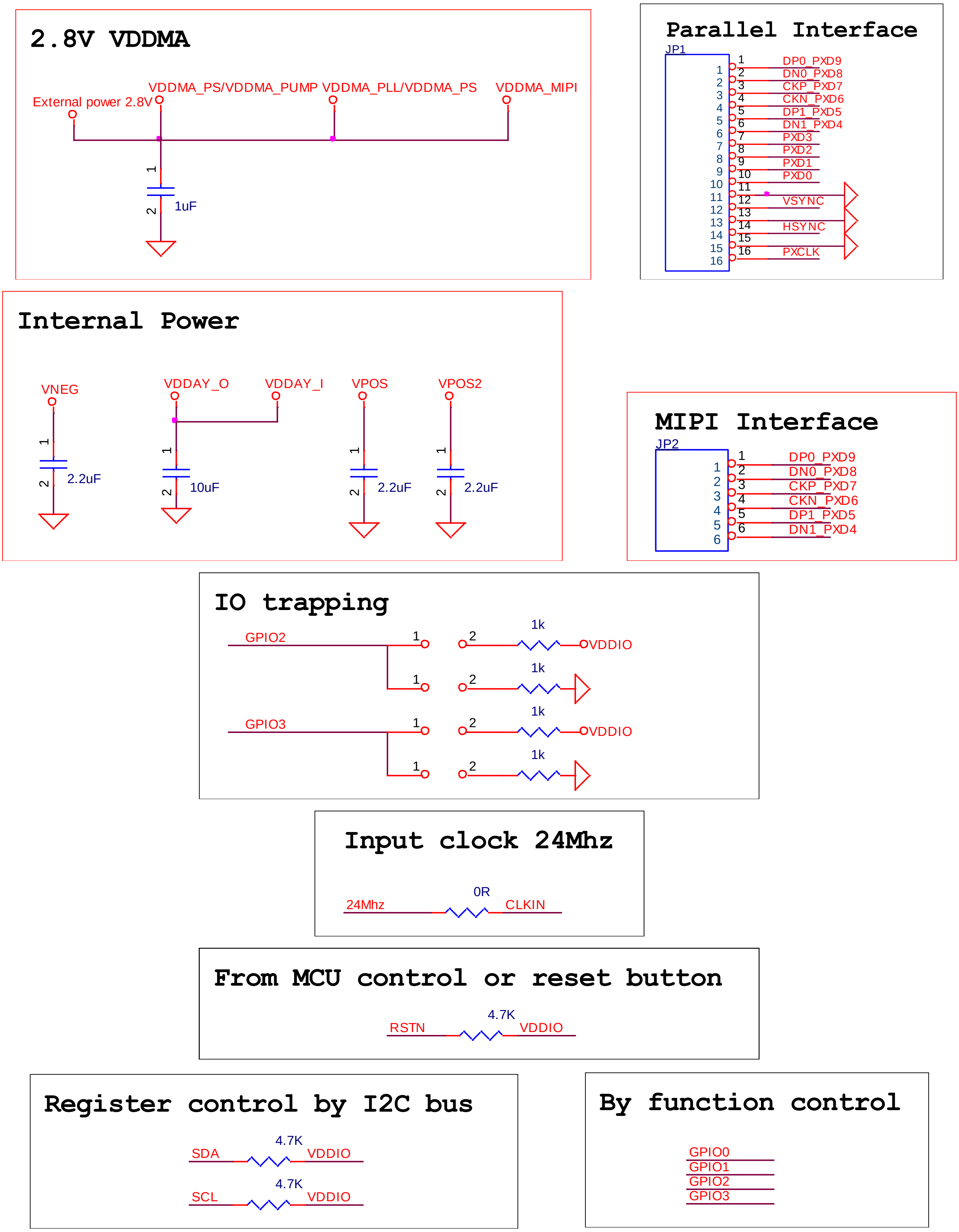



Figure 12. Reference Schematic Circuit

## 4.2 PCB Layout Design Guide

### 4.2.1 Design Rules

- Power input Capacitors:

1. Ceramic type capacitor is recommended.
2. Place the capacitor as close as possible to the sensor power pin
3. Avoid sharing the power supply design with other high-noise circuits.

- MIPI interface:

1. Isolate MIPI signals from other high-speed signals.
2. The differential pair trace lengths difference must be 5mil or less.
3. MIPI signals trace design on PCB board with either 100Ω differential impedance (± 20 %) or 50Ω impedance for each line (±15%).
4. MIPI traces must be routed on the same layer.
5. MIPI differential pair separation must be at least three times the width of a single trace width.

- Parallel interface:

Each trace separation must be at least three times the width of a single trace width.

- Pad size design:

1. If use solder mask defined (SMD), pad size refers to solder mask opening size.
2. If use non-solder mask defined (NSMD), pad size refers to copper metal size.

### 4.2.2 Recommend Footprint Dimension

Recommended dimension of pad is as shown in Figure 13 below.

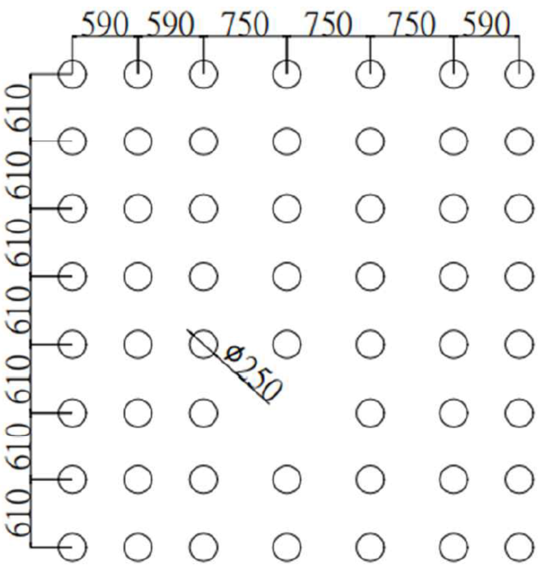



Figure 13. Recommend Footprint Dimension (Top view, unit: mm)

## 4.3 Assembly Guide

### 4.3.1 IR Reflow Solder Profile

Temperature profile is used for reflow soldering that must be fine-tuned to establish a robust process. The typical recommended IR reflow profile as shown below.

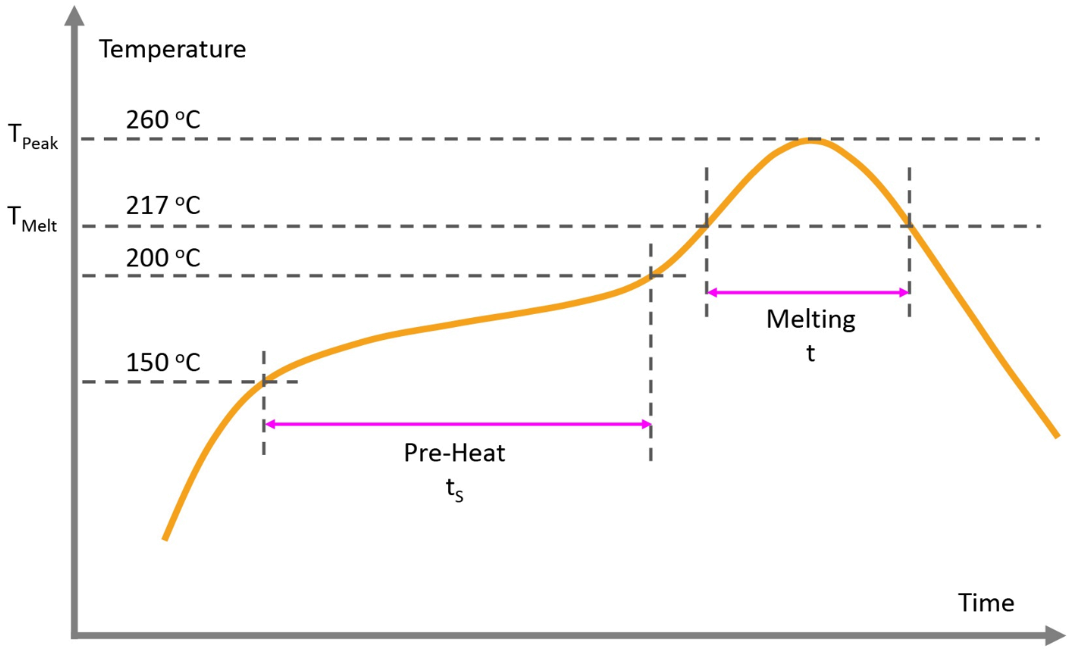



Figure 14. Reflow Profile

Table 11. Reflow Profile

| Parameter | Symbol | Min. | Max. | Unit | Note |
|---|---|---|---|---|---|
| Ramp-up rate | $T_{Ramp}$ | | 3 | °C/sec | From $T_{Melt}$ to $T_{Peak}$ |
| Pre-heat area temperature | $T_{Pre}$ | 150 | 200 | °C | |
| Pre-heat area duration | $t_S$ | 60 | 120 | sec | |
| Melting duration | t | 60 | 150 | sec | |
| Melting Temperature | $T_{Melt}$ | 217 | | °C | |
| Peak Temperature | $T_{Peak}$ | | 260 | °C | |

### 4.3.2 Under-filled Process

Epoxy under-filled process is required post IC mounting process as shown below.

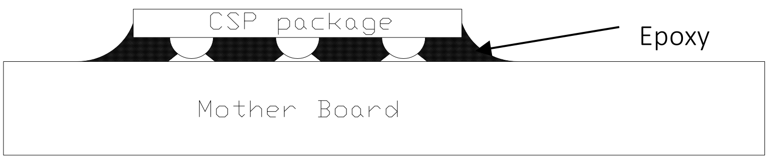



Figure 15. Epoxy Under-filled

## Revision History

| Revision Number | Date | Description |
|---|---|---|
| 0.5 | 01 Apr 2025 | Initial release |

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